

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Data Interpretation

Code of Conduct:

I _Shaik Mahaboob Basha (192365045)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mahaboob Basha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	
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Annexure A

Data Interpretation

Question 1: Hough Transform & Shape Detection

Scenario: Detect linear features (roads, rivers) in satellite images. **Input Data:**

- High-resolution grayscale satellite images of urban and rural areas (e.g., 1024 × 1024 pixels, TIFF or PNG)
- Images contain noise (clouds, shadows) and partial occlusions
- Multiple linear features (roads, rivers, canals) with intersections
- Optional synthetic dataset: images with drawn lines at random angles and lengths to test Hough Transform

Question 2: PCA & Shape Priors for Object Recognition

Scenario: Segment organs in MRI scans using shape priors. **Input Data:**

- MRI slices (grayscale, 512 × 512 pixels) of organs like liver or heart
- Training set: 50–100 annotated organ masks for PCA-based shape model construction
- Test set: 10–20 MRI images with partial organ boundaries or noise
- File formats: DICOM or PNG

Question 3: Machine Learning Methods (SVM, HMM, GMM, EM)

Scenario: Human activity recognition from video feature trajectories. **Input Data:**

- Video sequences (30–60 FPS) of activities like walking, running, jumping (e.g., 640 × 480 pixels, MP4)
- Feature trajectories extracted per frame: joint positions (x, y coordinates) or optical flow vectors
- Training labels: activity per video (e.g., walking, jumping)
- Optional synthetic data: CSV file with simulated joint coordinates over time

Question 4: Deep Learning (CNNs, Transformers, GANs)

Scenario: Detect rare obstacles in urban self-driving car images. **Input Data:**

- A. RGB camera images of urban streets (1024×768 pixels, JPG/PNG)
- B. Labeled dataset with bounding boxes for obstacles (rare objects, e.g., traffic cones, animals)
- C. Small dataset: 500–1000 labeled images for real obstacles
- D.

Optional synthetic images generated by GAN for data augmentation

Question 5: Graph-Based Learning for Shape Correspondence

Scenario: Find correspondences between 3D human face meshes. **Input Data:**

- A. 3D meshes of human faces (OBJ or PLY format) with varying vertex counts (5,000–50,000 vertices)
- B. Training set: 20–50 face meshes with annotated landmark correspondences
- C. Test set: 5–10 unseen face meshes
- D. Optional: Graph adjacency matrices representing each mesh for input to graph-based algorithms

Question 1: Hough Transform & Shape Detection

Scenario: Detect linear features such as roads, rivers and canals in high-resolution satellite images. The images contain noise, clouds, shadows, partial occlusions and intersections.

Answer:

The Hough Transform is a classical computer vision technique used to detect geometric shapes, particularly straight lines, even when lines are incomplete or partially occluded.

A suitable pipeline is:

Satellite Image → Grayscale Conversion → Noise Reduction → Edge Detection → Hough Transform → Line Detection → Shape Interpretation.

The image is first converted to grayscale. Gaussian or median filtering can reduce noise. Canny edge detection identifies strong edges. The standard Hough Transform represents a line as $\rho = x \cos \theta + y \sin \theta$. Edge points vote in Hough space, and peaks correspond to likely lines.

Noise can be reduced through filtering and suitable Canny/Hough thresholds. Partial occlusions can be handled because visible portions of the same line can vote for similar parameters. At intersections, multiple peaks can represent different line orientations.

Conclusion: The Hough Transform is useful for satellite-road and river detection because it can detect incomplete linear structures. Preprocessing and suitable thresholds are important to reduce false detections.

Question 2: PCA & Shape Priors for Object Recognition

Scenario: Segment organs such as the liver or heart in MRI scans using shape priors. The scenario specifies annotated organ masks for constructing a PCA shape model and test images with partial boundaries or noise.

Answer:

Principal Component Analysis (PCA) can learn the main variations in organ shape from annotated training masks.

Pipeline:

MRI Masks → Shape Alignment → Vectorization → Mean Shape → PCA → Shape Model → Test Image Segmentation.

The mean shape is calculated from training shapes, and PCA identifies the dominant modes of variation. A shape can be represented as $S = \bar{S} + Pb$, where $\bar{S}$ is the mean shape, P contains principal components and b contains shape parameters.

PCA shape priors can constrain segmentation toward anatomically plausible shapes when boundaries are incomplete. Gaussian filtering, normalization and other preprocessing methods can help with noise.

Advantages include dimensionality reduction, representation of common shape variations, recovery of missing boundaries and relatively simple computation. Limitations include dependence on representative training shapes and difficulty with highly unusual nonlinear deformations.

Conclusion: PCA-based shape priors are useful when organs have relatively consistent anatomical shapes and can improve robustness to noise and incomplete boundaries.

Question 3: Machine Learning Methods – SVM, HMM, GMM and EM

Scenario: Recognize human activities from video feature trajectories such as joint coordinates or optical-flow vectors. Activities include walking, running and jumping.

Answer:

Pipeline:

Video → Frame Extraction → Feature Extraction → Feature Trajectories → Machine Learning Model → Activity Classification.

SVM can classify activities using engineered features such as velocity, direction and joint displacement. HMM is particularly useful for sequential activities because it models temporal state transitions. GMM represents feature distributions as mixtures of Gaussian components. EM estimates GMM parameters by alternating between assigning probabilities to components and updating model parameters.

Comparison:

SVM – strong feature-based classification.

HMM – models temporal relationships.

GMM – models probability distributions.

EM – estimates latent GMM parameters.

Conclusion: HMM is especially suitable when temporal information is important, while SVM is effective for moderate-sized feature-based classification and GMM/EM can model motion distributions.

Question 4: Deep Learning – CNNs, Transformers and GANs

Scenario: Detect rare obstacles such as traffic cones and animals in urban self-driving-car images with limited labeled data.

Answer:

Pipeline:

RGB Image → Preprocessing → Object Detection Model → Feature Extraction
→ Bounding Box Prediction → Obstacle Classification.

CNNs learn spatial features such as edges, textures, shapes and object parts. Vision Transformers use image patches and attention to model relationships between regions and capture global context.

GANs can generate synthetic training images. The Generator creates images and the Discriminator distinguishes real from generated images. Synthetic examples can vary lighting, background, scale, orientation and position. GAN augmentation can increase data diversity and improve representation of rare classes, but generated images must be validated because unrealistic artifacts can harm training.

Conclusion: CNNs provide efficient feature extraction, Transformers provide strong global context, and GANs can increase training diversity when rare-object data is limited.

Question 5: Graph-Based Learning for Shape Correspondence

Scenario: Find correspondences between 3D human face meshes with varying vertex counts and annotated landmark correspondences.

Answer:

A 3D face mesh can be represented as a graph $G = (V, E)$, where V represents vertices and E represents connections between neighboring vertices.

Pipeline:

3D Face Mesh → Graph Construction → Feature Extraction → Graph Learning → Landmark Matching → Correspondence.

Features may include vertex coordinates, surface normals, curvature, local neighborhoods and geometric descriptors. Graph-based learning is flexible for irregular mesh structures and can learn relationships between facial regions such as the eyes, nose and mouth.

Advantages include suitability for irregular 3D data, use of local neighborhood relationships and ability to learn complex geometric patterns. Limitations include higher computational cost, dependence on graph construction and need for appropriate training data.

Conclusion: Graph-based learning is well suited to 3D face correspondence because facial meshes naturally form irregular graphs and may have different vertex densities.